

SLIIT
SE3082 – Parallel Computing



Lab 7
Exercise 05

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Exercise 05: Scatter + Allreduce

Implementation Code (sum_allreduce.c)

```
#include <mpi.h>
#include <stdio.h>
#include <stdlib.h>
#define N 1000000

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;

    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    int chunk_size = N / size;

    int *array = NULL;

    if (rank == 0) {
        array = (int *)malloc(N * sizeof(int));
        for (int i = 0; i < N; i++) array[i] = i + 1;
    }

    int *local_chunk = (int *)malloc(chunk_size * sizeof(int));

    MPI_Scatter(array, chunk_size, MPI_INT, local_chunk, chunk_size, MPI_INT, 0,
        MPI_COMM_WORLD);

    long long local_sum = 0;

    for (int i = 0; i < chunk_size; i++) local_sum += local_chunk[i];

    long long total_sum = 0;

    MPI_Allreduce(&local_sum, &total_sum, 1, MPI_LONG_LONG, MPI_SUM,
        MPI_COMM_WORLD);

    double percentage = ((double)local_sum / (double)total_sum) * 100.0;

    printf("Rank %d: local_sum = %lld, total_sum = %lld, contribution = %.2f%%\n", rank, local_sum,
        total_sum, percentage);

    if (rank == 0) free(array);

    free(local_chunk);
}
```

```
MPI_Finalize();  
return 0;  
}
```

Test Output (4 processes)

Rank 0: local_sum = 31250125000, total_sum = 500000500000, contribution = 6.25%

Rank 1: local_sum = 93750125000, total_sum = 500000500000, contribution = 18.75%

Rank 2: local_sum = 156250125000, total_sum = 500000500000, contribution = 31.25%

Rank 3: local_sum = 218750125000, total_sum = 500000500000, contribution = 43.75%